VICTORY 2027

The Circulatory System

LECTURE-7 BIOLOGY

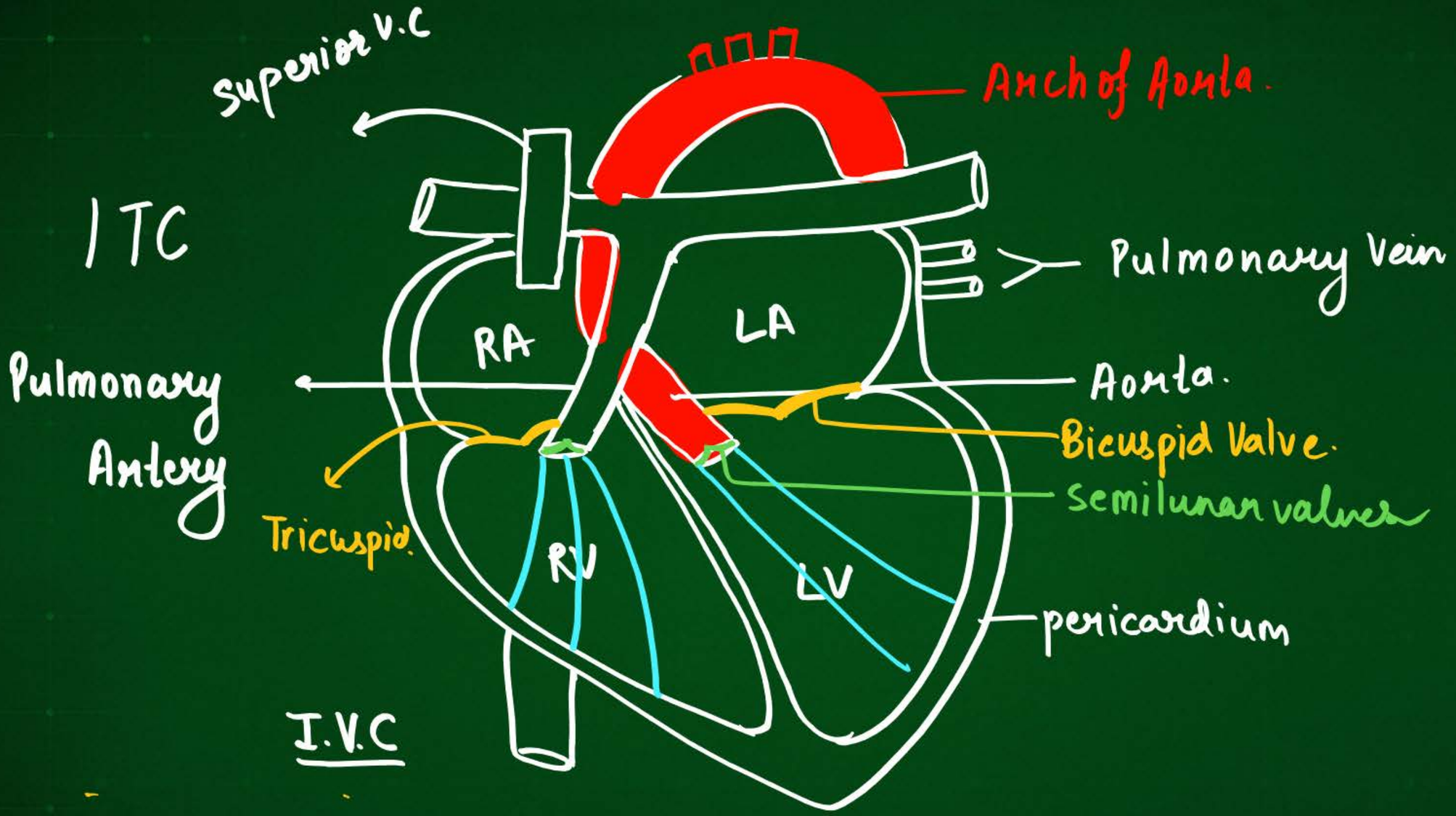
Upasana Ma'am



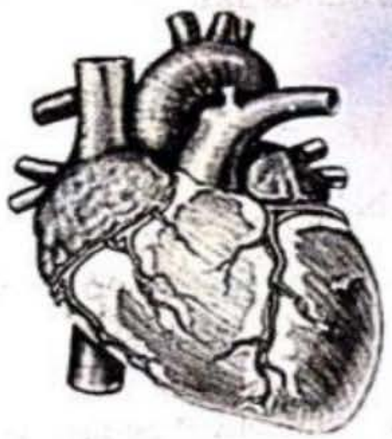
Topics To Be Covered

1. Human heart
2. Cardiac Cycle
3. Heart Disorders


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上


Superior V.C
Arch of Aorta.
ITC
Pulmonary Vein
RA
LA
Pulmonary Artery
Aorta.
Bicuspid Valve.
Semilunar valves
Tricuspid.
RV
LV
pericardium
I.V.C

(ii) Given below is the diagram of a human heart. Read the information below the diagram and fill in the blanks: [5]



2025

The human heart pumps blood throughout the body. It is the size of a large fist. The heart is located between the lungs in the thoracic cavity. It has four chambers. The heart functions all through the lifespan of a person and is responsible for the survival of the person.

5

The heart is enclosed by a membrane called (a) Pericardium The ventricles give rise to two large blood vessels called (b) P. Artery and (c) Aorta. The flaps of the cuspid valves are kept in position by (d) chordae tendenae (e) coronary Artery arteries supply oxygenated blood to the walls of the heart.

UMC uppu mummy dear

Q The first 'LUB' sound is mainly due to ?

A) Closing of Semilunar valves

B) Opening of Semilunar valves

C) Closing of AV valves ✓

D) Opening of AV valves

Q The second 'DUB' sound is mainly due to ?

A) Closing of Semilunar valves

B) Opening of Semilunar valves

C) Closing of AV valves

D) Opening of AV valves

Atrio - Ventricular valves

Q Which of the following are AV valves

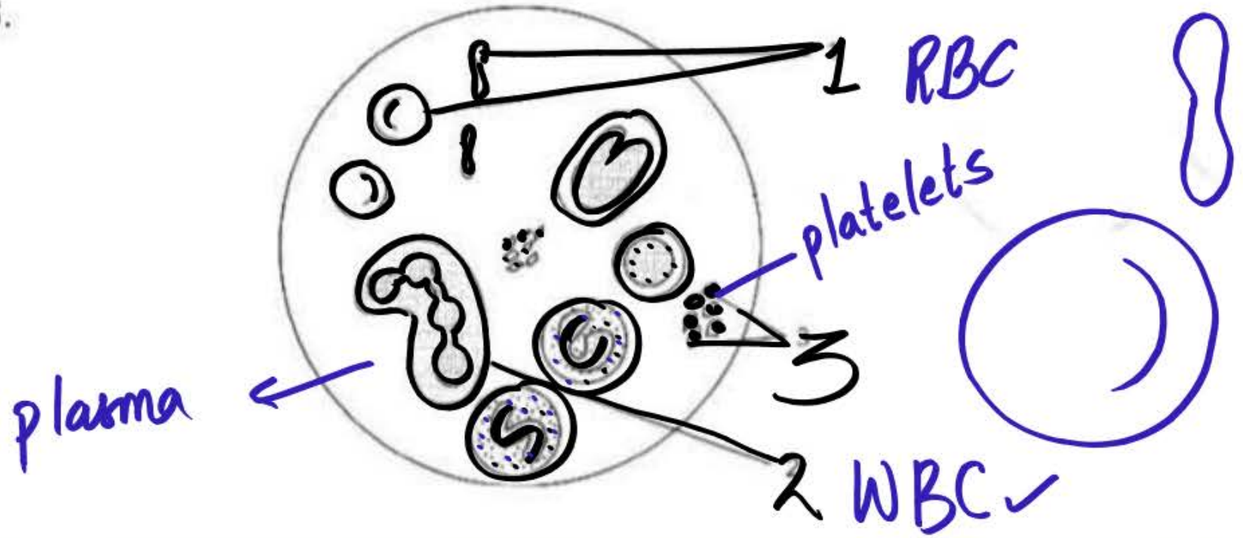
A) Tricuspid 3

B) Bicuspid

C) Mitral

D) All of the above

Given below is the diagram of a human blood smear. Label the parts numbered 1, 2 and 3.



V. Vimp.

'2027'
'DRAW'



platelets
enucleated.

RATTON KO KARO **BYE BYE !!**

EASY CONCEPTS SE KARO **HELLO HI !!**

With – *Uppu Ma'am*



CARDIAC CYCLE

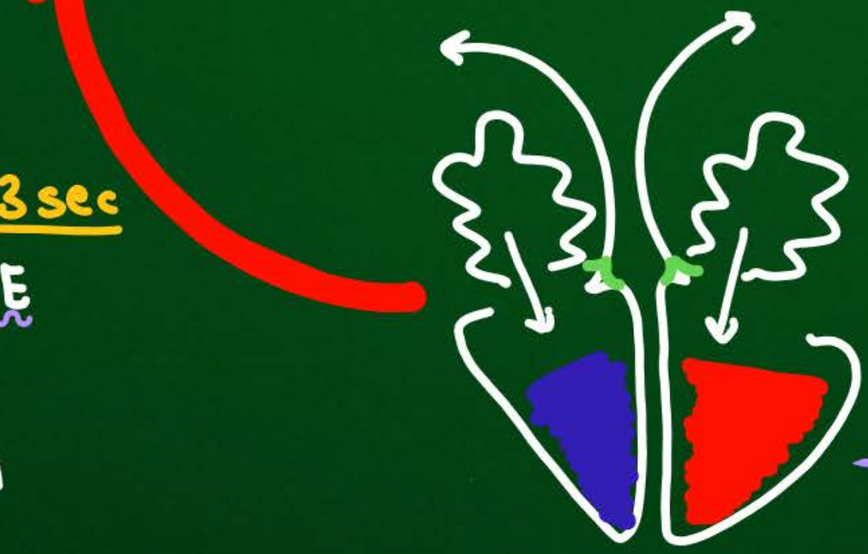
- The cardiac cycle is the complete sequence of events in one single heartbeat. It lasts about 0.8 seconds in a healthy human heart.

AVERAGE

75 times → 60 sec

1 → $\frac{60}{72-75}$ ⇒ 0.8 seconds

CARDIAC CYCLE



JOINT DIASTOLE [Both Atria & Ventricles Relaxed]

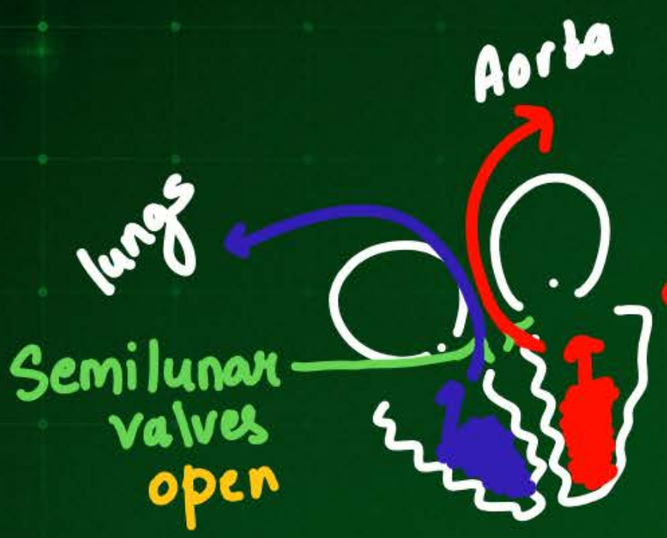
- Filling of both Atria.
- Oxygenated blood in L.A by P. Vein.
- Deoxygenated " " R.A by Vena Cava.
- A.V Valves OPEN ✓
- S.L Valves CLOSED 'DUB' sound

ATRIAL SYSTOLE (0.1 sec)

- Atrial Contraction
- A.V Valves Open
- S.L Closed.
- Ventricles are completely filled

VENTRICULAR SYSTOLE

- Atria Relaxing
- Ventricular Contraction 0.3 sec
- AV valves CLOSE 'LUB' sound
- SL valves open



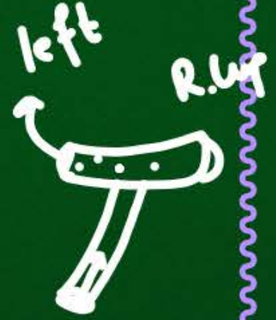


[0.4 sec]

'Heart Disorders'

Heart Attack / Myocardial Infraction





- Blood can clot even moving over rough surface.
- limited supply of O_2 will damage Cardiac Muscles.

Cardiac Arrest

- If there is no rhythmic contraction b/w the electrical signal sending tissue of heart, Sudden Cardiac Arrest can occur.

QUESTIONS

Which of the following has the thickest wall?

A. Right ventricle

B. Left ventricle

C. Right atium

D. Left atrium

Thank You